



Bank Marketing Prediction

Customer Subscription Prediction

Machine Learning Classification Project

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Introduction - Project Overview



What is the project?

A Machine Learning project that predicts whether a bank customer will subscribe to a marketing offer.



Why is it needed?

Banks contact many customers during campaigns. Prediction can help identify customers who are more likely to respond.



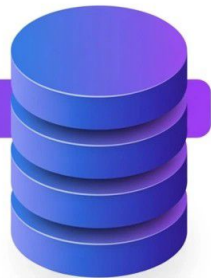
Learning approach

Supervised Learning → Classification → Binary Classification.



Main objective

Build and compare classification models using customer and campaign data.
Target: y | Outcome: Yes / No.



Dataset Source & Reference



UCI Machine Learning
Repository

Bank Marketing

 <https://archive.ics.uci.edu/dataset/222/bank+marketing>

 Public dataset

 13 Feb 2012

 Open Access CC BY 4.0



About Goal

Direct marketing Portuguese bank

Predict term deposit subscription • binary classification: yes / no



Dataset Info

- 45211 instances •
- 17 features •
- Classification •



Technical Details

- Feature Types: Categorical • Integer • Multivariate
- Size: 45k rows •
- Task: Classification •
- Used for benchmarking financial marketing



Dataset & Target Variable



Records

45,211



Columns

17



Dataset

Bank Marketing



Dataset Detail

UCI ML Repository

Source

Real-world marketing campaign data



Target Variable y



YES • Subscribed



NO • Not Subscribed

Binary Classification Problem • Supervised Learning • Two Outcomes



Machine Learning Workflow



The workflow prepares the raw data, balances target classes, selects useful features and finally compares classification models.



Data Preprocessing

5-Step Pipeline



PURPOSE — CLEAN & PREPARE DATA FOR RELIABLE ANALYSIS



Encoding & Class Balancing

Transforming categorical data and balancing target classes.

← Encoding

← LabelEncoder

← OneHotEncoder

← `pd.get_dummies`

- Convert categories → numeric
- Handle nominal features



Why SMOTE?

imbalanced classes

- Target class has far fewer samples
- Model biased toward majority class
- SMOTE generates synthetic samples



Class Distribution

Stage	No	Yes
Original	39922	5289
After SMOTE	39922	39922
Balanced	79844	—



Purpose: Prepare features via encoding, then use SMOTE to balance classes and improve model performance on the minority class.



Feature Selection & Transformation

Selecting key features and scaling for optimal modeling.



Purpose: Identify key features, normalize distributions, standardize scales, and prepare split dataset for robust model training



Classification Algorithm

Evaluating multiple classifiers for optimal performance.



Logistic Regression

linear model • probability



Decision Tree

decision rules • split



Random Forest

many trees • final prediction



AdaBoost

builds learners sequentially • corrects errors



Gradient Boosting

step-by-step • optimizing loss



PURPOSE: Compare accuracy | Assess robustness | Select the optimal classifier for deployment



Model Evaluation Results

Comparing model performance across key metrics

Model	Accuracy	Precision	F1	Recall
Logistic Regression	0.88	0.52	0.15	0.09
Decision Tree	0.84	0.32	0.35	0.35
Random Forest	0.89	0.55	0.35	0.25
AdaBoost	0.89	0.56	0.31	0.21
Gradient Boosting	0.89	0.56	0.32	0.22

Highest Accuracy

0.89

Random Forest + AdaBoost
+ Gradient Boosting

Highest Precision

0.56

AdaBoost + Gradient Boosting

Highest F1 / Recall

0.35 / 0.35

Decision Tree

Purpose: Evaluate models to select best for deployment.



How to Interpret the Results

Understanding evaluation metrics



Accuracy

How often the model is correct.

Confusion matrix and heatmap



Precision

Among customers predicted as positive, how many were actually positive



Recall

Among actual positive customers, how many were correctly identified



F1-score

A combined measure of Precision and Recall



Project observation

Random Forest, AdaBoost, and Gradient Boosting reach 0.89 accuracy. Decision Tree has highest F1 and Recall in notebook output. Therefore model selection should consider more than Accuracy alone.



Conclusion

Summary of achievements



What was achieved	Result:
Problem identified	Binary classification of bank customer response
Data preparation	Cleaning, encoding, outlier handling, and transformation
25 features selected	PowerTransformer
Feature selection	SelectKBest + ANOVA F-test
Models compared	5 classification algorithms
Best observed accuracy	0.89
Best observed F1 / Recall	Decision Tree — 0.33 F1 / 0.35 Recall

Purpose: Summarize modeling outcomes to guide next steps and deployment decision



Future Scope

1

Hyperparameter Tuning

Tune model parameters to improve performance

2

Improve Recall

Focus on identifying more actual positive customers

3

Feature Engineering

Create more useful customer campaign features

4

More Models

Experiment with additional classification techniques

5

Deployment

Build simple application for real-time customer prediction

Thank You